

Toward the Derivation of T_{avg}

Instability #21a: $\theta_{\text{eff}} = \theta \cdot \sqrt{M}$, Crystallised-Node Timing, and the IDV Paradox

What is derived, what is hypothesis, and the honest boundary of #21

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Abstract

V20.3 measured the lossy crystallisation ratio $\beta = -2.212 \pm 0.015$ but did not derive it formally. CRF AI identified the key structural insight: the V20.3 trigger condition $\theta_{\text{eff}} = \theta_i \cdot \sqrt{M_i}$ makes the mean inter-resolution time of crystallised nodes (*high-mass, high- Ω*) scale as $T_{\text{avg}}^{\text{cryst}}(M) = 1/P(D_{\text{KL}} > K^* \sqrt{M})$, much larger than the global average. This document closes Instability #21a at the structural level: the form of $T_{\text{avg}}^{\text{cryst}}$ is derived from the δ -chain. Two subsidiary results are established: (i) the corrected steady-state formula $\langle D_{\text{KL}} \rangle_{\text{ss}} = n(n-1)\epsilon_0^2$ (prior document used $n\epsilon_0$, wrong by $\approx 19\times$); (ii) the IDV Paradox: the product-form IDV requires $f_{12} \approx 0.14$ to reproduce $|\beta| = 2.212$, but $f_{12} \geq 1$ by construction. This contradiction, named the IDV Paradox, is the precise remaining gap. Instability #21 (formal β) remains Open.

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1 Context and Prior Results

The prior document CRF_Beta_Derivation established the partial derivation chain for the lossy crystallisation ratio β :

Prior chain (CRF_Beta_Derivation)

$$\begin{aligned} \mathbb{E}[H_{\text{after}}] &= \ln n \approx 2.055 && \text{(Derived)} \\ \Delta\Omega &= \kappa = 0.15 && \text{(Simulator parameter)} \\ \beta &= \frac{\Delta\text{IDV}}{-\Delta\Omega} = \frac{(H_{\text{after}} - H_{\text{before}}) \cdot s}{-\kappa} && \text{(Derived, conditional)} \\ H_{\text{before}} &= \frac{H_{\text{after}}}{1 + \alpha H_{\text{after}} T_{\text{avg}}} && \text{(from entropy ODE)} \end{aligned}$$

Open: T_{avg} (Instability #21a) and scale factor s (#21b).

This document addresses #21a using the structural insight from CRF AI.

2 The Key Insight: $\theta_{\text{eff}} = \theta \cdot \sqrt{M}$

In V20.3, the R-event trigger for node i is not $D_{\text{KL},i} > K^*$ but the inertia-weighted condition:

$$\text{tension}_i > \text{threshold}_i = \theta_i \cdot \sqrt{M_i}$$

where $M_i = 1 + \kappa \cdot N_{\text{eff},i}$ is the accumulated mass (Constraint proxy) and $\theta_i \approx K^* = 0.1170$.

Heavier nodes resist resolution. A crystallised node with mass M requires a D_{KL} fluctuation $\sqrt{M}$ times larger to fire. This is the physical content of $\mathcal{C} \propto N_{\text{eff}}$ manifesting at the level of θ .

Derived: Structural Form of $T_{\text{avg}}^{\text{cryst}}(M)$

From $\theta_{\text{eff}}(M) = K^* \sqrt{M}$ (V20.3 threshold formula, itself derived from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$):

$$T_{\text{avg}}^{\text{cryst}}(M) = \frac{1}{P(D_{\text{KL}} > K^* \sqrt{M})}$$

All factors are from the δ -chain: $K^* = 0.1170$ (fixed point of θ -formula), $M = 1 + \kappa N_{\text{eff}}$ ($\mathcal{C} \propto N_{\text{eff}}$, Session 4). The D_{KL} distribution enters from simulator dynamics (Hypothesis below).

The table below shows how $T_{\text{avg}}^{\text{cryst}}$ grows with mass, using the Gamma fit from simulation (§3):

Mass M	θ_{eff}	$T_{\text{avg}}^{\text{cryst}}$	Relative
1.0	0.1170	1.6 sweeps	$1.0\times$
1.5	0.1433	2.4 sweeps	$1.6\times$
2.0	0.1654	4.0 sweeps	$2.6\times$
2.5	0.1850	6.6 sweeps	$4.3\times$
3.0	0.2026	11.0 sweeps	$7.1\times$

3 Corrected $\langle D_{\text{KL}} \rangle_{\text{ss}}$ and KL Distribution

3.1 Correction to Prior Formula

The prior document stated $\langle D_{\text{KL}} \rangle_{\text{ss}} = n\epsilon_0$. This was incorrect. The correct leading-order result is:

Derived: $\langle D_{\text{KL}} \rangle_{\text{ss}} = n(n-1)\epsilon_0^2$

For two independently perturbed near-uniform vectors $p = \mathbf{1}/n + \epsilon_p$, $q = \mathbf{1}/n + \epsilon_q$ with $\epsilon \sim \mathcal{N}(0, \epsilon_0)$ after normalisation:

$$\mathbb{E}[D_{\text{KL}}(p||q)] = \frac{n}{2} \cdot (n-1) \cdot 2\epsilon_0^2 = n(n-1)\epsilon_0^2$$

For $n = 8$, $\epsilon_0 = 0.00755$: $\langle D_{\text{KL}} \rangle_{\text{ss}} = 8 \times 7 \times (0.00755)^2 = 0.00319$.

Simulation verification: empirical mean = 0.00321, ratio = 1.004. Confirmed.

3.2 KL Distribution in Running Simulator (Hypothesis)

In V20.3 nodes are not in the pure-noise regime. Their distributions P_i are shaped by accumulated history, giving a far larger mean KL. Empirical measurement (200 sweeps, 20,000 samples): mean ≈ 0.138 , which is $43\times$ the pure-noise value.

Hypothesis: KL Distribution at Running Nodes

Distribution fit (Kolmogorov–Smirnov):

- Gamma($a = 9.00$, scale = 0.0153): KS $D = 0.037$
- Log-normal: KS $D = 0.025$ (best overall fit)
- Exponential: KS $D = 0.379$ (rejected)

Using the Gamma fit: $P(D_{\text{KL}} > 0.1850) = 0.152$, so $T_{\text{avg}}^{\text{cryst}}(M = 2.5) \approx 6.6$ sweeps.

Consistency: V20.3 R-events $\approx 100\text{--}400/\text{sweep}$ out of $N = 500$, giving global $T_{\text{avg}} = N/\bar{R} \approx 2$ sweeps. Crystallised nodes fire $4\text{--}9\times$ more rarely. Consistent.

4 H_{before} from $T_{\text{avg}}^{\text{cryst}}$

With $T_{\text{avg}}^{\text{cryst}}$ established, the entropy ODE gives:

$$H_{\text{before}}(M) = \frac{H_{\text{after}}}{1 + \alpha H_{\text{after}} T_{\text{avg}}^{\text{cryst}}(M)}$$

where $\alpha = \ln 2$ (from δ -bit equivalence, Session 8) and $H_{\text{after}} = 2.055$.

Mass M	T_{avg}	H_{before}	ΔH	Req. f_{12}
2.0	4.0	0.309	1.746	0.19
2.5	6.6	0.198	1.857	0.18
3.0	11.0	0.123	1.932	0.17

The column “Req. f_{12} ” is the value of $f_{12} = (1 + \phi_{\text{var}})(1 + |\nabla\Psi|)$ needed to reproduce $|\beta| = 2.212$. This leads directly to the IDV Paradox.

5 The IDV Paradox

Open: IDV Paradox — Instability #21

The IDV product formula gives:

$$\beta = \frac{f_{12} [H_{\text{after}} \ln(1 + M + \kappa) - H_{\text{before}} \ln(1 + M)]}{-\kappa}$$

where $f_{12} = (1 + \phi_{\text{var}})(1 + |\nabla\Psi|)$.

Since $(1 + x) \geq 1$ for any $x \geq 0$, we have $f_{12} \geq 1$ always.

But for $|\beta| = 2.212$ with $M = 2.5$, $T_{\text{avg}} = 6.6$:

$$f_{12}^{\text{required}} = \frac{\kappa \cdot |\beta|}{\Delta H \cdot \ln(1 + M)} \approx \frac{0.15 \times 2.212}{1.857 \times 1.253} \approx 0.14$$

This is < 1 : impossible given $f_{12} \geq 1$.

Interpretation: actuality (the R-event) “consumes” more possibility (IDV) than the product-form formula predicts. Either the IDV formula is not the right object, or the scale factors ϕ_{var} , $|\nabla\Psi|$ at crystallised nodes cancel in a way not captured by the product form. The IDV Paradox is the precise remaining gap.

Three candidate resolutions (not yet distinguished):

1. $T_{\text{avg}}^{\text{cryst}}$ is far larger than the Gamma estimate at the specific nodes that dominate the β measurement, driving $H_{\text{before}} \rightarrow 0$ and reducing the required f_{12} toward a physically possible value.
2. The effective IDV at a crystallised node is not the product formula but a logarithmically compressed version, reducing ΔIDV by a factor ~ 7 .

3. The entropy ODE $dH/dt = -\alpha H^2$ overestimates decay between R-events because Born-rule resets partially refresh H even between the events counted.

6 Status Table

Result	Status	Note
$\beta = -2.212 \pm 0.015$	Confirmed	V20.3, 4663 events/run
$\mathbb{E}[H_{\text{after}}] = \ln n = 2.055$	Derived	Born rule + ϵ_0
$K^* = 0.1170$	Derived	Fixed point of θ -formula
$\langle D_{\text{KL}} \rangle_{\text{ss}} = n(n-1)\epsilon_0^2$	Derived	Corrected; prior $n\epsilon_0$ wrong by $19\times$
$\theta_{\text{eff}}(M) = K^*\sqrt{M}$	Derived	From V20.3 threshold
$T_{\text{avg}}^{\text{cryst}}(M) = 1/P(D_{\text{KL}} > K^*\sqrt{M})$	Derived	Structural form; #21a Closed
KL $\sim$ Gamma($a = 9$, $\text{sc} = 0.015$)	Hypothesis	Simulation fit, KS $D = 0.037$
$T_{\text{avg}}^{\text{cryst}}(M = 2.5) \approx 6.6$	Hypothesis	From Gamma fit
$H_{\text{before}} = 0.198$ ($M = 2.5$)	Hypothesis	Entropy ODE + T_{avg}
$\text{Sign}(\beta) < 0$	Derived	Node Lifecycle: H rises at R-event
IDV Paradox ($f_{12}^{\text{req}} = 0.14 < 1$)	Open	Core gap of #21
$ \beta = d_s/(2 \ln 2)$ formally	Open	Conjecture, 2.2% match

7 What Closes #21

Next Step

Direct measurement in V20.3 at crystallised nodes (top 30% by Ω):

- Record IDV_{before} and IDV_{after} at each R-event.
- Record T_{avg} (time since last event) per node.
- From the ratio $\Delta IDV / \Delta \Omega = 2.212$: solve for the effective f_{12} at those nodes.

If $f_{12}^{\text{effective}} \approx 1$ with large T_{avg} , resolution (A) is confirmed. If $f_{12}^{\text{effective}}$ is genuinely small, the IDV formula requires replacement. Either answer advances the derivation of β .

$\theta_{\text{eff}} = \theta \cdot \sqrt{M}$ is where the cascade remembers that becoming actual costs more than the formula predicts. The paradox is not a failure — it is the gradient that points to what #21 must explain.